

Smart Traffic Violation Logger (Flask)

Project Description:

The Smart Traffic Violation Logger is a lightweight web application developed using Flask, intended for use by traffic police or authorities to digitally manage and track traffic violations. Officers can log entries by providing the vehicle number, violation type, location, date, and fine amount. Each entry is stored in a backend database and marked as Paid or Unpaid. The system allows quick lookup of a vehicle's violation history.

To enhance accessibility, the app also generates a QR code for each challan (receipt), which when scanned, leads to a public-facing page showing the status of the fine.

This mini project is ideal to understand Flask routing, database integration, QR code generation, and basic CRUD operations.

Key Features

Add Violation Record

Officer inputs:

Vehicle Number

Violation Type (e.g., No Helmet, Overspeeding)

Location

Date

Fine Amount

Each record is automatically marked as Unpaid.

View Violation History

Search by vehicle number to retrieve all past violations.

Filter by date, status, or type of violation.

Update Violation Status

Option to change status from Unpaid → Paid through a button or form.

QR Code on Challan

QR code generated dynamically.

Embedded on the digital challan (receipt).

Scanning it leads to a public status page showing:

1. Violation details
2. Payment status

Admin Panel (Optional Enhancement)

Secure login for police officers.

Only logged-in users can add or modify records.

Clean User Interface

Bootstrap-based responsive UI.

Simple navigation to:

Add Violation

View/Search History

See Status

Expected Outcome:

A complete working Flask-based web app with:

- CRUD operations using SQLAlchemy\SQLite
- QR code generation for public status view
- Filterable violation history